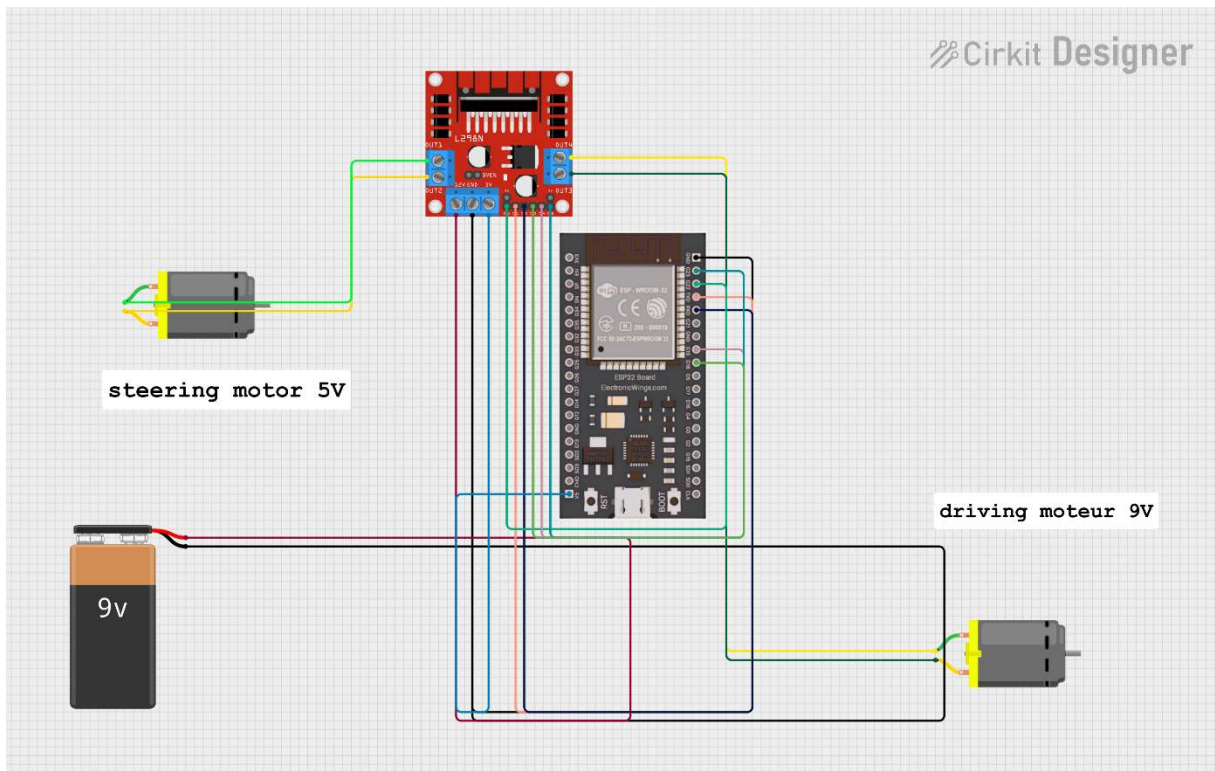


Project documentation

1. Components

1. ESP 32 microcontroller and the dabble application via bluetooth
2. Two DC motors: one for driving and one for steering
3. L298N motor driver
4. Old RC car wheels and structure

2. Diagram



sorry about the confusing diagram

component	Connected to esp32	notes
ENA (L298N)	GPIO 22	PWM for Drive Motor
IN1 (L298N)	GPIO 16	Motor forward/reverse
IN2 (L298N)	GPIO 17	Motor forward/reverse
ENB (L298N)	GPIO 23	PWM for Steering Motor
IN3 (L298N)	GPIO 18	Motor direction
IN4 (L298N)	GPIO 19	Motor direction
L298N GND	ESP32 GND	Must be connected!
L298N +12V	7V battery (+)	Supply for motors

3. Main Functions:

1. rotateMotor(rightSpeed, leftSpeed) - Controls the direction and speed of the right (drive) and left (steering) motors using digital pins and PWM.
2. setUpPinModes() - Initializes motor driver pins as outputs and sets up PWM channels using ledc functions.

2. `setup()` - Called once on boot. It initializes all pins and starts the BLE connection with the name "MyBluetoothCar".
3. `loop()` - Continuously checks for GamePad input via Dabble. - Handles drive (Up/Down), steering (Left/Right), and full stop (Triangle) commands.
Button Mappings: - Up (0): Drive forward - Down (1): Drive backward - Left (2): Steer left - Right (3): Steer right - Triangle (4): Stop all motors All speed values are limited to defined constants to ensure control and protect components.

4.Code

Although I faced many difficulties by using old PWN code so I had to install the 2.0.17 so I can use functions I can't explain any more because I do not have any more space to write in but I will gladly answer any questions

```
rccartourna.ino
1  #define CUSTOM_SETTINGS
2  #define INCLUDE_GAMEPAD_MODULE
3  #include <DabbleESP32.h>
4  #include "Arduino.h"
5  const int enableRightMotor = 22;
6  const int rightMotorPin1 = 16;
7  const int rightMotorPin2 = 17;
8  const int enableLeftMotor = 23;
9  const int leftMotorPin1 = 18;
10 const int leftMotorPin2 = 19;
11 #define MAX_DRIVE_SPEED 255
12 #define MAX_STEERING_SPEED 140
13 const int PWMfreq = 1000;
14 const int PWMResolution = 8;
15 const int rightMotorPWMSpeedChannel = 4;
16 const int leftMotorPWMSpeedChannel = 5;
17
18 void rotateMotor(int rightMotorSpeed, int leftMotorSpeed) {
19     if (rightMotorSpeed < 0) {
20         digitalWrite(rightMotorPin1, LOW);
21         digitalWrite(rightMotorPin2, HIGH);
22     } else if (rightMotorSpeed > 0) {
23         digitalWrite(rightMotorPin1, HIGH);
24         digitalWrite(rightMotorPin2, LOW);
25     } else {
26         digitalWrite(rightMotorPin1, LOW);
27         digitalWrite(rightMotorPin2, LOW);
28     }
29
30     if (leftMotorSpeed < 0) {
31         digitalWrite(leftMotorPin1, LOW);
32         digitalWrite(leftMotorPin2, HIGH);
33     } else if (leftMotorSpeed > 0) {
34         digitalWrite(leftMotorPin1, HIGH);
35         digitalWrite(leftMotorPin2, LOW);
36     } else {
37         digitalWrite(leftMotorPin1, LOW);
38         digitalWrite(leftMotorPin2, LOW);
39     }
40
41     ledcWrite(rightMotorPWMSpeedChannel, abs(rightMotorSpeed));
42     ledcWrite(leftMotorPWMSpeedChannel, abs(leftMotorSpeed));
43 }
44
45 void setupPinModes() {
46     pinMode(enableRightMotor, OUTPUT);
47     pinMode(rightMotorPin1, OUTPUT);
48     pinMode(rightMotorPin2, OUTPUT);
49     pinMode(enableLeftMotor, OUTPUT);
50     pinMode(leftMotorPin1, OUTPUT);
51     pinMode(leftMotorPin2, OUTPUT);
52     ledcSetup(rightMotorPWMSpeedChannel, PWMfreq, PWMResolution);
53     ledcSetup(leftMotorPWMSpeedChannel, PWMfreq, PWMResolution);
54     ledAttachPin(enableRightMotor, rightMotorPWMSpeedChannel);
55     ledAttachPin(enableLeftMotor, leftMotorPWMSpeedChannel);
56
57     rotateMotor(0, 0);
58 }
59
60 void setup() {
61     setupPinModes();
62     Dabble.begin("MyBluetoothCar");
63 }
64
```

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